

**NATIONAL JUNIOR COLLEGE
SH 2 PRELIMINARY EXAMINATION**

Higher 1

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

Paper 2 Structured Questions

8873/02

**Tues 21 Aug 2018
2 hours**

Candidates answer **all** questions on the Question Paper.
Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your subject class, registration number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs.

Do not use staples, paper clips, glue or correction fluid/tape.

Section A

Answer **all** the questions.

Section B

Answer **only one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use			
1	/5		
2	/10		
3	/7		
4	/12		
5	/16		
6	/8		
7	/5		
8 OR 9	/20		
Penalty units			
Penalty sf			
Paper 2	/83		

Answer ALL questions on the space provided.

This paper consists of **20** printed pages.

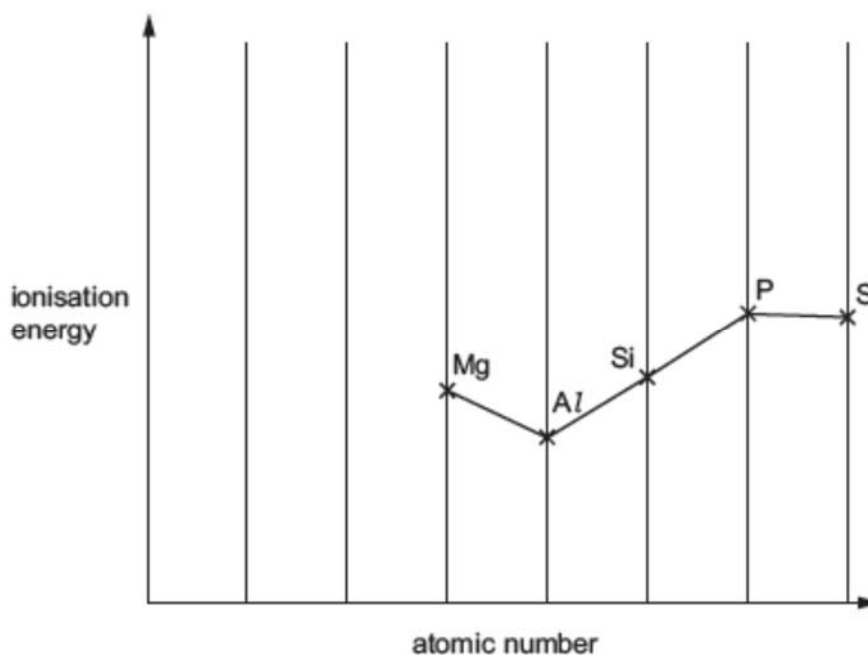
Section A

Answer **all** the questions in this section, in the spaces provided.

- 1 The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements.

- (a) The elements in the third period, and their compounds, show trends in their physical and chemical properties.

A sketch graph of the first ionisation energies of five successive elements in the third period is shown.



- (i) Sketch on the graph, the position of the ionisation energy of the two elements that come before Mg in this sequence. [1]
- (ii) Explain, with reference to electronic arrangements, the decreases in first ionisation energy between Mg and Al and between P and S.

Mg and Al:

.....

.....

P and S:

.....

.....

[2]

- (b) The chlorides of the elements in the third period behave in different ways when added to water, depending on their structure and bonding.

L is a chloride of an element in Period 3. A student investigated L and the results are as given below.

- L is a white crystalline solid with a melting point of 987 K.
- L dissolves in water to form a weakly acidic solution.
- Addition of NaOH(aq) to an aqueous solution of L produces a white precipitate.

- (i) Identify L.

L: [1]

- (ii) Write an equation to illustrate the formation of the weakly acidic solution.

..... [1]

[Total: 5]

- 2 Nanomaterials research takes a materials science-based approach to nanotechnology, leveraging advances in materials metrology and synthesis which have been developed in support of microfabrication research. Materials with structures at the nanoscale often have unique optical, electronic, or mechanical properties as compared to its corresponding bulk materials.

Graphene is a nanomaterials which is an allotrope of carbon such as graphite and diamond. It has many uncommon properties. Graphene is the strongest material ever tested, efficiently conducts heat and electricity, and is nearly transparent.

Because nanotechnology is a recent development, the health and safety effects of exposures to nanomaterials and nanoparticles, and what levels of exposure may be acceptable, are subjects to ongoing research.

- (a) State the difference between a nanoparticle and a nanomaterials in terms of size.

.....
 [1]

- (b) Describe the property of nanomaterials. Explain how this property enables it to be used in the following application:

1 Catalysis

.....

.....

.....

2 Climbing ability of geckos

.....

.....

.....[4]

- (c) With the aid of a labelled diagram, describe the structure of graphene. Explain how this structure gives graphene the properties of high electrical conductivity and high tensile strength.

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.....[3]

- (d) Cerium oxide is being used in electronics, biomedical supplies, energy and fuel additives.

Predict how nanoparticles of cerium oxide could present a risk to human health **and** the environment.

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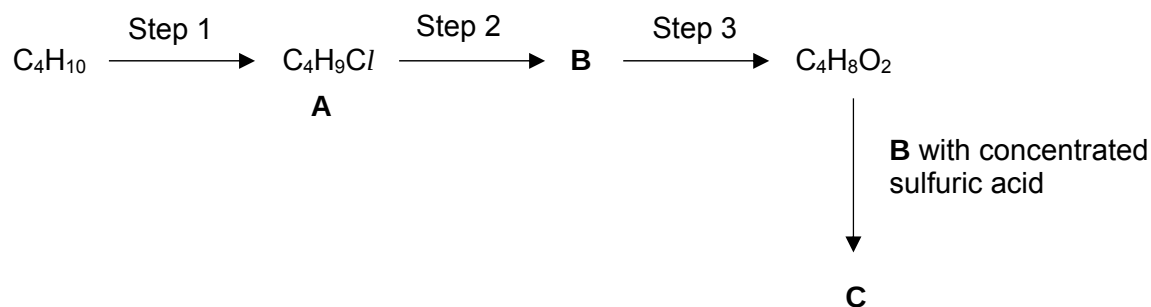
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.....[2]

[Total: 10]

- 3 Student **M** suggested a 2-step process as shown, to synthesise **B** from a suitable butane.



- (a) Identify the structures of compounds **B** and **C**. Suggest reagents and conditions for each of the steps.

Structure of B	Structure of C
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Step 1:

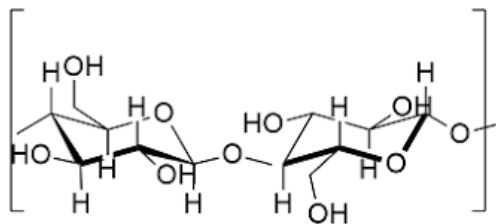
Step 2:

Step 3: [5]

- (b) Compound **D**, an isomer of compound **A**, is formed after **step 1**. Suggest the structure of compound **D**. Hence, state the approximate ratio in which compounds **A** and **D** are formed.

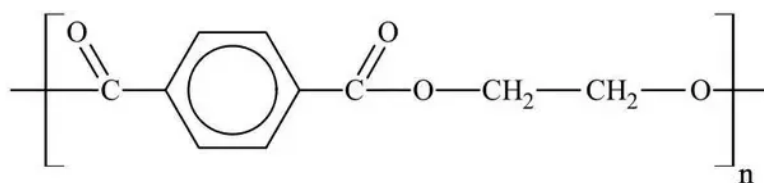
[2]
[Total:7]

- 4 In the past, football shirts were made from cotton, or even wool. These cotton or wool shirts trapped heat and readily soaked up any sweat produced, making them rather uncomfortable to wear.



Repeating unit of cotton

Nowadays, polyethylene terephthalate (PET) and spandex are used in manufacture of football shirts. Cotton can absorb 7% of its weight in water, whereas polyester only absorbs about 0.4% of its weight. PET also does not crease easily.



Repeating unit of PET

- (a) (i) Suggest why cotton can absorb more water than PET.

.....
 [1]

- (ii) Explain why PET does not crease easily with respect to its **structure**.

.....
 [1]

- (b) (i) Calculate the percentage of mass of carbon in PET.

[2]

- (ii) Identify the type of polymerisation in the formation of PET and provide **two** reasons to explain your answer.

Type of polymer

Reason 1

.....

Reason 2

.....

[3]

- (iii) Explain why a PET fabric dissolves slowly when placed in aqueous alkali.

.....

.....

[1]

- (iv) Write an equation for **one repeat unit** of PET reacting with aqueous sodium hydroxide.

[1]

- (c) PET is a thermoplastic. With respect to the structure and interactions of polymer chains, explain

- (i) the difference between thermoplastics and thermosets.

.....

.....

[1]

- (ii) why PET can be used as a fabric.

.....

.....

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[2]

[Total: 12]

- 5 On 11 May 2018, Mount Merapi on Central Java, Indonesia erupted, causing the local airport to be closed. The eruption was reported to be caused by accumulation of volcanic gases. The volcanic ash and gases spewed can be dangerous to planes passing through the plume. The most abundant volcanic gas is harmless water vapour. However, significant amounts of carbon dioxide, sulfur dioxide, hydrogen sulfide and hydrogen halides are also emitted.

When carbon dioxide is emitted from volcanoes, it typically becomes diluted to low concentrations very quickly and is not life threatening. However, cold carbon dioxide gas can flow into low-lying areas where it can reach much higher concentrations. Breathing air with more than 3% CO₂ can quickly lead to headaches, dizziness, increased heart rate and difficulty breathing. At about 15%, unconsciousness and death can result quickly.

Table 1: Volcanic gas composition in area A

Gas	Volume Percentage
Water vapour, H ₂ O	87.1
Carbon dioxide, CO ₂	unknown
Sulfur dioxide, SO ₂	0.5
Hydrogen, H ₂	0.7
Carbon monoxide, CO	0.01
Hydrogen sulfide, H ₂ S	0.23
Hydrogen halides	unknown

Composition of the volcanic gases are typically expressed in terms of volume percentage, which can be calculated as follows

$$\text{volume percentage} = \frac{\text{volume of gas}}{\text{total volume}} \times 100\%$$

- (a) (i) Draw the dot-and-cross diagram of H₂S.

[1]

- (ii) State and explain whether H₂S or H₂O has a larger bond angle with respect to its central atom.

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.....

[2]

- (iii) Suggest a reason why at low temperature, carbon dioxide would sink rapidly and accumulate to high concentrations.

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[2]

- (iv) People living in area **A** were evacuated as the level of carbon dioxide was increasing rapidly. Given that 0.3 g of CO₂ was present in 1 dm³ of gas mixture at r.t.p, determine the volume percentage of carbon dioxide present.

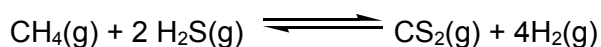
Hence comment on the possible danger if people remained in the area.

Volume percentage of carbon dioxide =

.....

[2]

- (b) Hydrogen sulfide can react with methane in the following equation.



1 mol of CH₄, 2 mol of H₂S, 1 mol of CS₂ and 1 mol of H₂ were allowed to reach equilibrium in a 2 dm³ container at a constant temperature and pressure.

- (i) State the oxidation state of carbon in each carbon-containing compound.

CH₄ : CS₂ : [1]

- (ii) Hence comment on the role of H₂S in the reaction.

.....
 . [1]

- (iii) Given that CS_2 was found to be 1.2 mol at equilibrium, determine the value of K_c , giving its units.

[2]

- (iv) Using the *Data Booklet*, calculate the enthalpy change of the above reaction. Assume bond energy of C-S bond is 272 kJ mol^{-1} .

[2]

- (v) Hence state and explain the effect of increasing the temperature of the system on
- position of the equilibrium
 - rate of the reaction.

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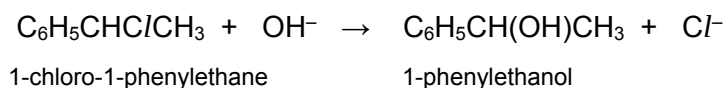
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[3]

[Total: 16]

- 6 1-chloro-1-phenylethane undergoes hydrolysis with hydroxide ions to produce 1-phenylethanol, as shown in the equation below.



The rate of this reaction can be studied by measuring the amount of hydroxide ions that remain in the solution at a given time after the reaction has been quenched.

- (a) Suggest a suitable quenching agent that can be used to slow down the rate of the reaction effectively.

..... [1]

- (b) The rate of this reaction was measured using different initial concentrations of the two reagents and the results are shown below.

Experiment	$[\text{C}_6\text{H}_5\text{CHClCH}_3] / \text{mol dm}^{-3}$	$[\text{OH}^-] / \text{mol dm}^{-3}$	Relative rate
1	0.05	0.10	1.0
2	0.10	0.20	2.0
3	0.15	0.10	3.0

- (i) Deduce the order of reaction with respect to each of the reagents. Explain your reasoning.

Order with respect to $\text{C}_6\text{H}_5\text{CHClCH}_3 = \dots\dots\dots$

Order with respect to $\text{OH}^- = \dots\dots\dots$ [2]

- (ii) Write the rate equation for this reaction, stating the units of the rate constant, k .

rate = $\text{mol dm}^{-3} \text{s}^{-1}$

units of $k = \dots\dots\dots$

[2]

[Total: 5]

- 7 Lithium is a scavenger for hydrogen, hence able to prevent the reaction between hydrogen and copper during pure copper casting. Reaction with hydrogen makes the copper brittle, causing it to fall apart under very light stress.

Table 2

	Enthalpy / kJ mol ⁻¹
Enthalpy change of formation of LiAlH ₄ (l)	-152.5
Enthalpy change of formation of LiAlH ₆ (s)	-454

- (a) Heating lithium in a stream of hydrogen gas produces white, crystalline, ionic lithium hydride, LiH.

- (i) By quoting relevant data from the *Data Booklet*, suggest and explain how the magnitude of the lattice energy of LiCl would compare to that of LiH.

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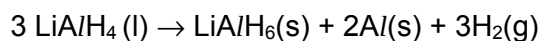
 [2]

- (b) When lithium hydride is heated with anhydrous aluminium chloride, lithium aluminium hydride, LiAlH₄ and lithium chloride are produced.

Balance the equation for the above reaction.

..... LiH + AlCl₃ → LiAlH₄ + LiCl [1]

- (c) Just above its melting point, LiAlH₄ decomposes according to the following equation.



Using **Table 2**, calculate the standard enthalpy change of this reaction.

[2]
 [Total: 5]

Section B

Answer only **one** question from this section, in the spaces provided.

- 8(a)** With the aid of a Boltzman distribution curve, explain how the presence of a catalyst increase the rate of a reaction.

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[3]

- (b)** The melting points of some compounds are given.

Substance	Octan-1-ol	Iodine	Fullerene	Graphite
Formula	$\text{CH}_3(\text{CH}_2)_9\text{OH}$	I_2	C_{60}	C
Melting point/ K	277	286	873	>3000

- (i)** Explain why the melting points of octan-1-ol and iodine are comparable.

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[2]

- (ii)** Explain why the melting point of graphite is higher than that of fullerene.

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[2]

- (iii) By quoting relevant data from the *Data Booklet*, account for the trend in the thermal stabilities from HCl to HI .

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 [2]

- (iv) Arrange the pH of HCl to HI in aqueous solution in ascending order.
 Explain the variation you have outlined.

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 [2]

- (c) Compound **X** has the molecular formula $\text{C}_4\text{H}_8\text{O}_2$.

- (i) Treatment of **X** with acidified potassium dichromate(VII) produces a green solution.
 What could be the functional groups present in **X**?

.....
 [1]

- (ii) There is no reaction when **X** is treated with aqueous sodium carbonate, Na_2CO_3 .
 What could be the functional groups absent in **X**?

.....
 [1]

- (iii) When **X** is shaken with aqueous bromine, the orange colour disappears.
 What could be the functional groups present in **X**?

.....
 [1]

(iv) The molecule of **X** has the following features.

- The carbon chain is unbranched and the molecule is not cyclic.
- No oxygen atom is attached to any carbon atom which is involved in π bonding.
- No carbon atom has more than one oxygen atom joined to it.

There are four possible isomers of **X** which fit these data.

Draw displayed formulae of **all** possible isomers and state which type of isomerism they show.

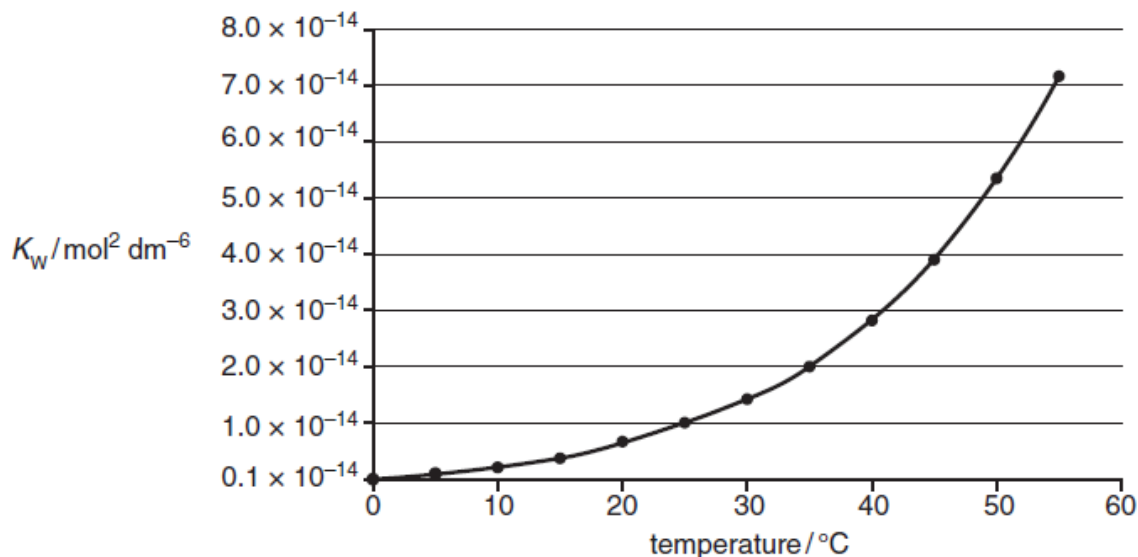
[6]
[Total: 20]

- 9 (a) The dissociation of water is a reversible reaction.



The ionic product of water, K_w , measures the extent of dissociation of water. K_w varies with temperature. It is always important to quote the temperature at which measurements are being taken.

The graph below shows the variation of K_w between 0 °C and 60 °C.



- (i) Write the expression for K_w .

.....

[1]

- (ii) Calculate the concentration of hydroxide ions, $[\text{OH}^-]$, in an aqueous solution of hydrochloric acid at pH of 4.37 at 25 °C.

[2]

- (iii) Using the graph, explain whether the dissociation of water is an exothermic or endothermic process.

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.....

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[2]

- (iv) Using the graph, determine the pH of pure water at body temperature, at 37 °C.

[3]

- (v) Given the following data:

Indicator	Colour change (acidic to basic medium)	pH range in which colour change occurs
Bromothymol blue	yellow to blue	6.0 – 7.6

State the colour observed when a few drops of bromothymol blue is added to a sample of pure water at body temperature, at 37 °C.

.....

[1]

- (vi) Buffer solutions are important in biological systems and in industry to maintain almost constant pH values

In the human body, one important buffer system in blood involves the hydrogencarbonate ion, HCO_3^- and carbonic acid, H_2CO_3 , which is formed when carbon dioxide dissolves in water.

Use the following equation to explain how this buffer maintains a constant pH of 7.41 even if a small amount of acid enters the bloodstream.



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[2]

- (b) The hydrocarbons **A**, C_4H_{10} , and **B**, C_4H_8 , are both unbranched.
A does **not** decolourise bromine.
B decolourises bromine and shows cis-trans isomerism.

- (i) Draw the skeletal formula of **A**.

[1]

- (ii) The hydrocarbon **A**, C_4H_{10} , has a branched isomer.
 Suggest why unbranched **A** has a higher boiling point than its branched isomer.

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[2]

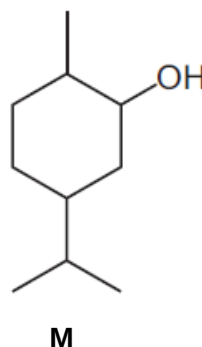
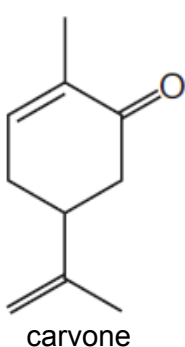
- (iii) Draw the structure of the product obtained from the reaction between **B** and bromine. State the type of reaction that occurred.

Type of reaction:

Product:

[2]

- (c) Carvone occurs in spearmint and can be converted to compound **M**.



- (i) State the reagent and conditions for the conversion of carvone to compound **M**.

name of reagent.....

conditions.....

[1]

- (ii) Draw the structure of the product when **M** is treated with excess concentrated sulfuric acid at 170 °C.

[1]

- (d) Write an equation for the reaction that occurs when silicon tetrachloride is added to water. Predict the pH of the resulting solution.

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[2]

[Total: 20]

End of Paper